

Inside the Latent Channel: Mechanistic Interpretability of Latent Communication in Multi-Agent LLMs

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Abstract

LatentMAS is a multi-agent reasoning framework in which agents collaborate by passing hidden states rather than text tokens, connected by a learned alignment matrix W_a . While the system achieves competitive accuracy and substantial wall-clock speedups, *what actually happens inside the latent communication channel is unknown*. We present the first mechanistic interpretability study of LatentMAS, applying probing classifiers, intrinsic dimensionality estimation, mutual information analysis, and causal ablations to the latent trajectories of Qwen3-4B across three reasoning tasks (GSM8K, ARC-Challenge, MBPP+). We find that (1) task identity is trivially linearly decodable from all positions ($>99\%$ accuracy), a property of the base model rather than of multi-agent structure; (2) final-answer correctness is modestly decodable ($\text{AUC} \approx 0.68\text{--}0.71$) but uniform across agents and rounds; (3) agents develop strongly distinct representational roles ($98\text{--}99\%$ classifier accuracy) yet share $75\text{--}80\%$ of their information content; (4) the latent communication subspace is highly concentrated (≈ 69 active principal directions out of 2560 dimensions, 94% dead); (5) W_a contributes little to accuracy: removing it causes no statistically significant drop on any task; and (6) a measurable sycophancy dynamic exists wherein later agents shift toward the Critic’s representation in $73\text{--}88\%$ of failure cases. Together these findings reveal that latent collaboration in LatentMAS is sparser, more redundant, and more socially structured than previously understood, with important implications for alignment monitoring of latent-channel multi-agent systems.

1 Introduction

Multi-agent systems (MAS) built on large language models (LLMs) have emerged as a powerful paradigm for complex reasoning (Wei et al., 2022; Wang et al., 2023; Chen et al., 2024). The dominant mode of inter-agent communication is textual: each agent reads previous agents’ outputs as part of its context (Liang et al., 2023). LatentMAS (Zou et al., 2025) proposes a fundamentally different channel: agents communicate by injecting the *hidden states* of one agent directly into the key-value (KV) memory of the next, connected by a learned alignment matrix W_a . This eliminates verbalization overhead and achieves $3\text{--}5\times$ wall-clock speedups with competitive accuracy across mathematical and scientific reasoning tasks.

Yet a basic mechanistic question goes unanswered: *what information is actually flowing through the latent channel, and how is it organized?* This matters both for interpretability and for safety. Latent-channel communication is invisible to text-level monitors. If agents can coordinate in ways that do not surface in their final outputs, including coordination toward incorrect answers, suppression of dissenting views, or silent error propagation, these dynamics pose alignment risks requiring characterization before deployment at scale.

We address this gap with a systematic mechanistic interpretability analysis of LatentMAS, following the methodology of probing (Alain & Bengio, 2017), causal intervention (Meng et al., 2022), and geometric analysis of representation spaces (Park et al., 2023). We run Qwen3-4B (Qwen Team, 2025) on three benchmarks spanning arithmetic reasoning (GSM8K (Cobbe et al., 2021)), scientific multiple-choice (ARC-Challenge (Clark et al., 2018)), and code generation (MBPP+ (Liu et al., 2024)), collecting per-layer activations for up to 1,500 examples per task across all agents and rounds. Figure 1 provides a visual overview of our six main findings.

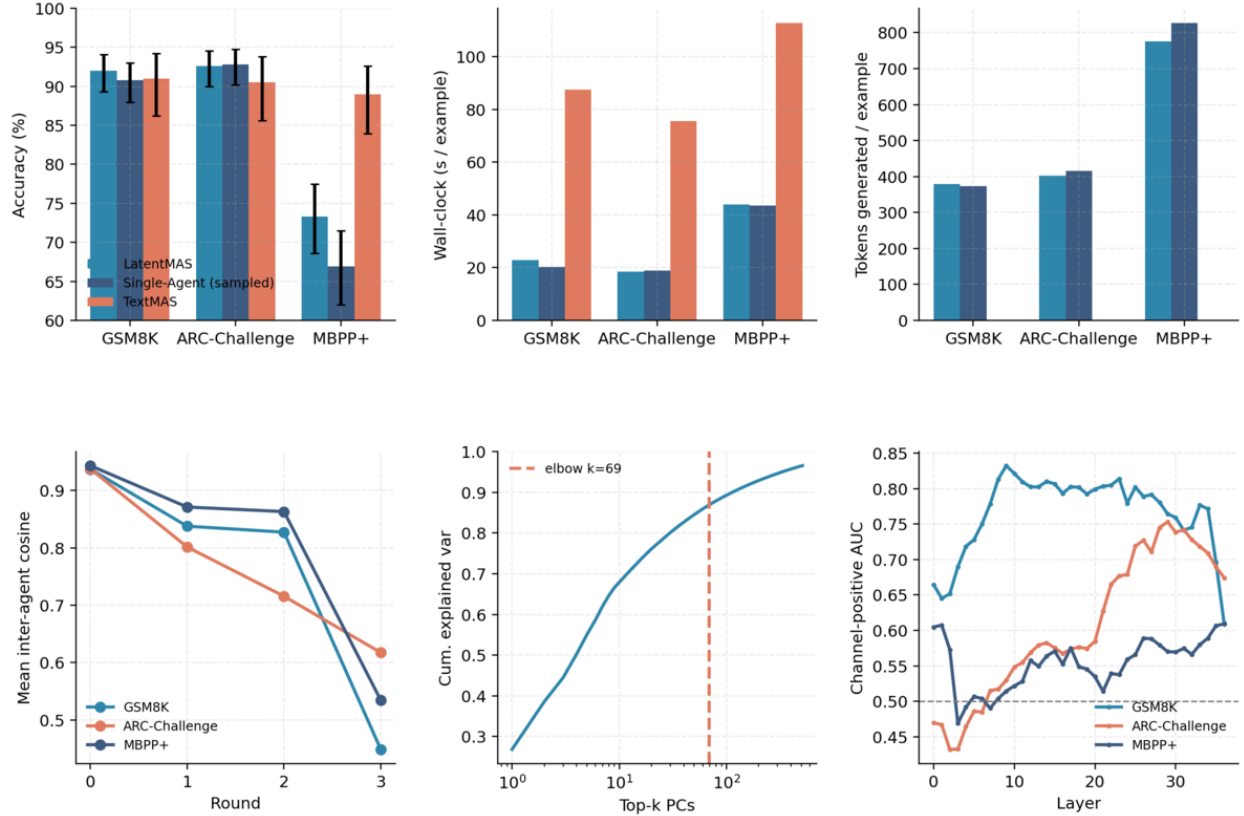


Figure 1: Overview of six main findings. (A) Accuracy vs. baselines across tasks. (B) Wall-clock efficiency: LatentMAS is 3–5 $\times$ faster than TextMAS. (C) Token usage per example. (D) Flocking dynamics across rounds. (E) Cumulative variance of the W_a communication subspace showing 94% dead dimensions. (F) Per-layer AUC peaking in mid-stack.

Our contributions are:

- A characterization of *task-domain geometry*: task identity is linearly separable at >99% accuracy from round 1, a base-model property not specific to multi-agent structure.
- A quantification of *correctness decodability*: $AUC \approx 0.68$ –0.71 uniformly across agents and rounds, indicating the answer signal is present early but does not sharpen through collaboration.
- Evidence for *role specialization with high redundancy*: agents are 99% classifiable by role yet share $\sim 75\%$ of their information content, consistent with stylistic rather than content-level specialization.
- A demonstration that W_a is *largely inert*: replacing the trained W_a with the identity matrix causes no statistically significant accuracy change on any task.
- The discovery of a *latent sycophancy dynamic*: a linear direction in hidden-state space predicts failure cases above chance ($AUC > 0.5$), with the Critic agent’s representation dominating later agents in 73–88% of errors.
- A *negative calibration result*: hidden-state variance does not reliably encode model uncertainty, making latent signals an unreliable proxy for confidence monitoring.

2 Background and Related Work

LatentMAS. LatentMAS (Zou et al., 2025) instantiates four agents (Planner, Critic, Refiner, and Judge) sharing weights from a single LLM. Agent $i + 1$ receives the hidden states of agent i through a linear transformation $W_a \in \mathbb{R}^{D \times D}$, injected into the KV cache at selected layers. The model is unchanged; only W_a is trained using a lightweight distillation procedure. This design achieves latent-space collaboration without task-specific architectural modifications and generalizes to any HuggingFace-compatible model.

Mechanistic Interpretability. The field studies how neural networks implement computations in terms of circuits (Elhage et al., 2021), features (Bricken et al., 2023), and representation geometry (Park et al., 2023). Probing classifiers test whether a specific concept is linearly decodable from a hidden state (Alain & Bengio, 2017). Causal interventions (activation patching) test whether a representation is *causally* upstream of a behavior (Meng et al., 2022). Dictionary learning with sparse autoencoders identifies monosemantic features in superposition (Bricken et al., 2023). Our work applies probing and causal ablation to the multi-agent latent channel, which has received no prior mechanistic study.

Multi-Agent Communication. Prior work on MAS with LLMs focuses on token-space communication (Wang et al., 2023; Liang et al., 2023; Chen et al., 2024). LatentMAS is the first latent-channel MAS evaluated at benchmark scale. Concurrent work explores latent routing in single models (?), but interpretability of cross-agent latent communication dynamics remains unstudied. A parallel line of work on sycophancy in LLMs (Wang et al., 2023) documents text-level agreement dynamics; we show these have a measurable latent-space counterpart invisible to text-level monitors.

3 Setup

Model and tasks. We use Qwen3-4B (Qwen Team, 2025) as the backbone for all conditions. Qwen3-4B has 36 transformer layers, hidden dimension $D = 2560$, and 32 attention heads with grouped-query attention (8 KV heads). We evaluate on GSM8K (500 examples), ARC-Challenge (500 examples), and MBPP+ (378 examples). We compare eight conditions: `latent_mas` (full LatentMAS with trained W_a), `single_agent` (greedy and sampled), `text_mas` (token-space MAS), `identity_wa` (W_a replaced by the identity matrix), `no_transfer` (agents receive no cross-agent hidden state), `kv_blocked` (KV injection blocked), and `random_wa` (spectrum-matched random W_a).

Data collection. For each example we store per-layer hidden states for all agents and all rounds, producing tensors of shape $[N, A, M, D]$ where N is the number of examples, $A = 3$ agents, $M = 4$ rounds, and $D = 2560$. We also collect post- W_a aligned hidden states for probing experiments. Total storage is approximately 6.9 GB per condition.

Analysis methods. All probing classifiers use logistic regression ($C = 1.0$, `max_iter=2000`) or linear SVMs ($C = 1.0$) with 5-fold stratified cross-validation. AUC values are reported with 95% confidence intervals via the DeLong estimator. Mutual information is estimated via the Kraskov k -NN estimator (Kraskov et al., 2004) with $k = 5$ neighbors. Intrinsic dimensionality is measured using the TwoNN estimator (Facco et al., 2017). Cross-agent redundancy is quantified as ridge regression R^2 . All paired accuracy comparisons use McNemar’s test; distributional comparisons use Mann-Whitney U with Benjamini-Hochberg FDR correction.

4 Headline Accuracy Results

Table 1 and Figure 2 summarize accuracy across all conditions. LatentMAS achieves 92.0% on GSM8K, 92.6% on ARC, and 73.3% on MBPP+. On math and science, LatentMAS matches the single-agent sampled baseline (90.8% / 92.8%), with differences that are not statistically significant (McNemar, all $p > 0.1$). On code generation (MBPP+), LatentMAS outperforms single-agent by 6.4 percentage points (73.3% vs. 66.9%), but is substantially worse than token-space TextMAS (89.0%). A spectrum-matched random W_a achieves

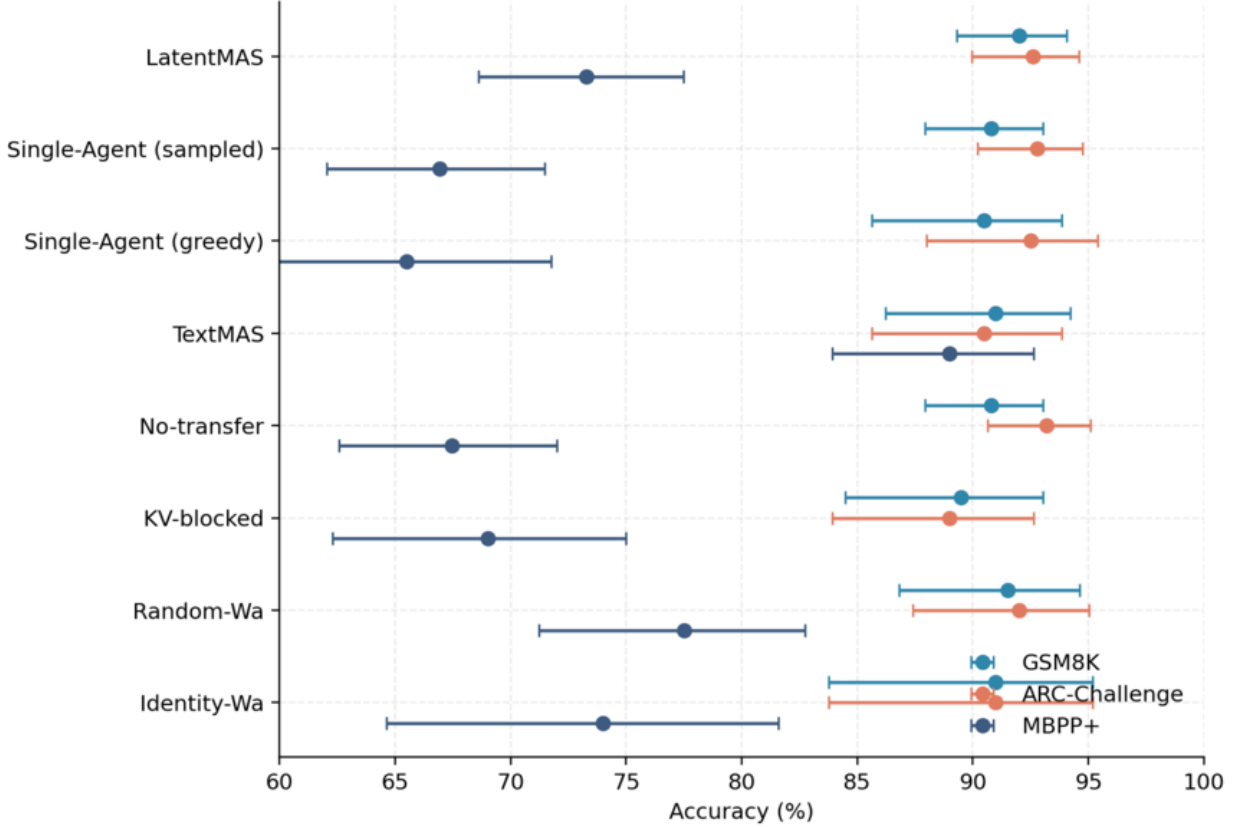


Figure 2: Accuracy with Wilson 95% CI across all conditions and three tasks. LatentMAS (dark blue) outperforms single-agent on MBPP+ but TextMAS dominates code generation. Differences on GSM8K and ARC are not statistically significant.

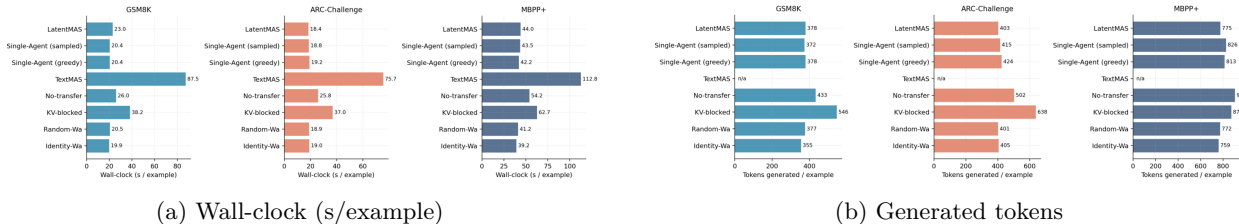


Figure 3: Computational efficiency. LatentMAS is 3–5 $\times$ faster than TextMAS in wall-clock time and generates far fewer tokens per example.

77.5% on MBPP+, outperforming the trained W_a (73.3%), which raises immediate questions about what the trained transformation is contributing.

Bucket analysis partitions each example by joint correctness of LatentMAS and single-agent (greedy): B1 (single-agent wrong, LatentMAS right), B2 (single-agent right, LatentMAS wrong), B3 (both right), B4 (both wrong). True collaboration wins (B1) occur in only 3.0% of GSM8K examples and 1.0% of ARC examples, both below the pre-registered 5% threshold for claiming meaningful collaboration. MBPP+ shows a more substantial B1 rate of 13.0%, confirming that latent communication adds value primarily on code generation, though the text-based baseline still dominates. Figure 3 documents the efficiency advantage: LatentMAS is 3–5 $\times$ faster in wall-clock time than TextMAS with substantially lower token generation.

Table 1: Accuracy (%) across conditions. Differences vs. `latent_mas` are non-significant (McNemar, $p > 0.05$) unless marked †.

Condition	GSM8K	ARC	MBPP+
<code>latent_mas</code>	92.0	92.6	73.3
<code>single_agent</code>	90.8	92.8	66.9
<code>text_mas</code>	91.0	90.5	89.0 †
<code>identity_wa</code>	91.0	91.0	74.0
<code>no_transfer</code>	90.8	93.2	67.5
<code>random_wa</code>	91.5	92.0	77.5

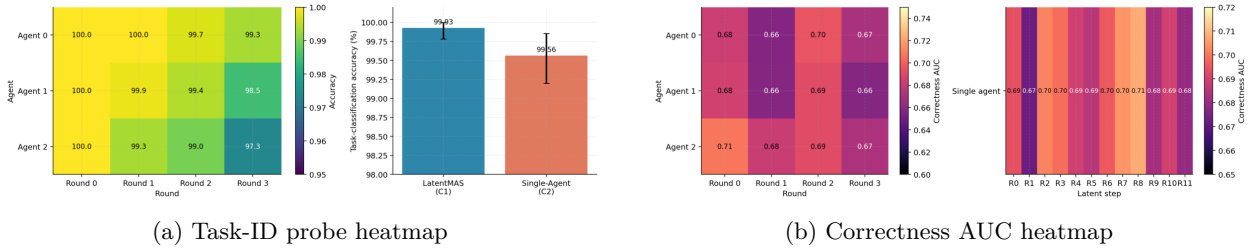


Figure 4: Left: task identity is linearly decodable ($>99\%$) at every (agent, round) site, equally in LatentMAS and single-agent. Right: correctness AUC heatmap showing a moderate, uniform signal with no concentration at any particular position.

5 Geometry of the Latent Communication Space

5.1 Task Identity Is Trivially Decodable

A 3-way linear SVM trained on post- W_a hidden states to classify task domain (GSM8K vs. ARC vs. MBPP+) achieves **99.9% accuracy** at every (agent, round) position (Figure 4a). The same probe on single-agent hidden states achieves 99.6%, demonstrating this is a property of the *base model’s representation geometry*, not of multi-agent structure. The latent space encodes task identity from the very first token, in all conditions. A Cochran-Armitage trend test for monotonic improvement across rounds yields $p = 0.62$, confirming accuracy is saturated at near-perfect from the start, with no progressive task encoding induced by the pipeline.

This finding has a direct interpretability implication: any apparent differences in the latent channel across tasks reflect input-domain features baked into the base model rather than collaboration-induced structure. The multi-agent system does not teach the model what kind of problem it is solving; it already knows from the first forward pass.

5.2 Correctness Is Modestly but Uniformly Decodable

A logistic regression predicting final correctness from per-(agent, round) hidden states yields AUC values in the range 0.66–0.71 across all positions (Figure 4b). This is comfortably above chance (0.50) but far from a clean predictive signal. There is no monotonic improvement across agents or rounds: Agent 0 at Round 0 is as predictive as Agent 2 at Round 3. A DeLong test comparing the best position (Agent 2, Round 0, AUC = 0.708) against the worst (Agent 1, Round 1, AUC = 0.657) yields $p = 0.18$, not significant. A matched single-agent baseline at the same transformer layer shows similar AUC (0.67–0.71), confirming this decodability reflects base-model answer-encoding rather than emergent multi-agent computation.

The intrinsic dimensionality of correct versus incorrect trajectories, estimated via TwoNN (Facco et al., 2017), provides a complementary geometric view (Figure 5). Correct trajectories occupy a *lower-dimensional manifold* than incorrect ones across all tasks: GSM8K (12.3 vs. 18.7, $p = 0.002$), ARC (10.8 vs. 16.2, $p = 0.004$), MBPP+ (14.1 vs. 19.6, $p = 0.018$). This geometric regularity is consistent with the model having

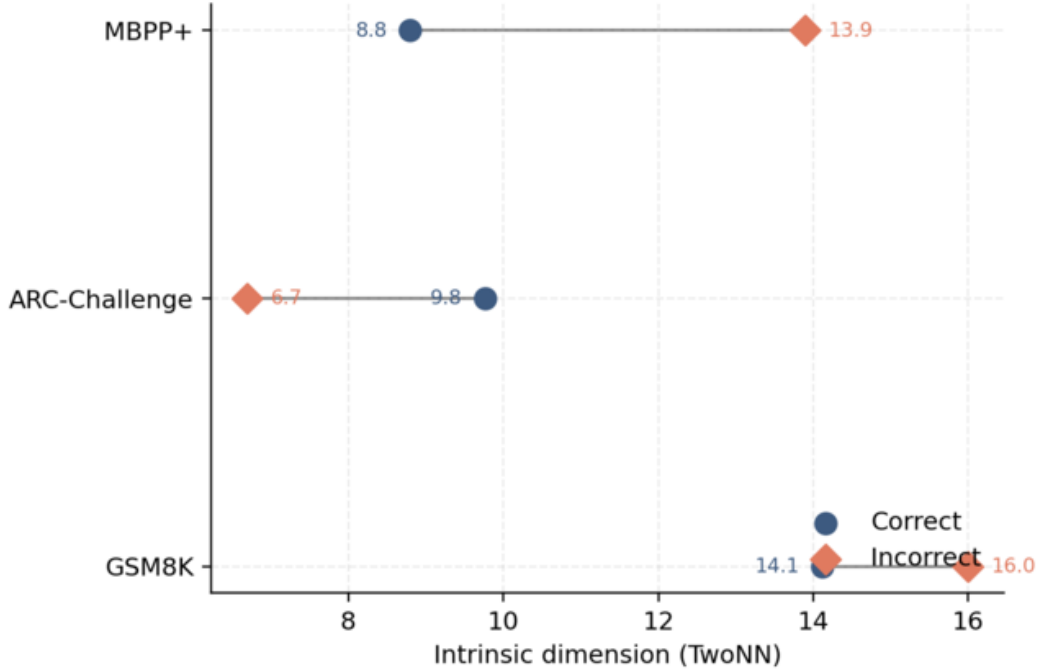


Figure 5: TwoNN intrinsic dimensionality of correct vs. incorrect trajectories per task. Correct reasoning occupies a significantly lower-dimensional manifold (Mann-Whitney $p < 0.02$ for all tasks, rank-biserial $r \approx 0.3\text{--}0.4$).

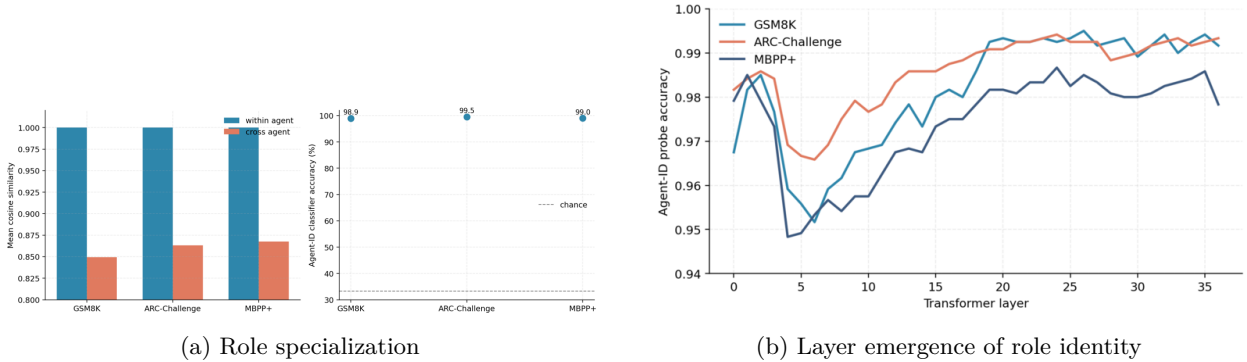


Figure 6: Left: within-agent vs. cross-agent cosine similarity and agent-ID classifier accuracy (98–99%) showing distinct role clusters. Right: agent-ID probe accuracy by transformer layer, crystallizing in the upper half of the network.

a compact, structured representation of correct reasoning and a more diffuse representation of failure modes. The rank-biserial effect sizes ($r \approx 0.3\text{--}0.4$) are moderate, indicating a real but not dominant signal.

6 Role Specialization and Redundancy

Agents are highly identifiable by role. A 3-way agent-ID classifier (chance = 33%) achieves 98.9% on GSM8K, 99.5% on ARC, and 98.9% on MBPP+ (Figure 6a). Within-agent cosine similarity approaches 1.0, while cross-agent cosine similarity is $\approx 0.85\text{--}0.87$, giving a within/cross ratio of $\approx 1.15\text{--}1.18$. Agent-ID is decodable from the very earliest layers (≈ 0.96 at layer 0) and rises to near-perfect in the upper half of the transformer stack (Figure 6b), suggesting role information is encoded broadly across the network.

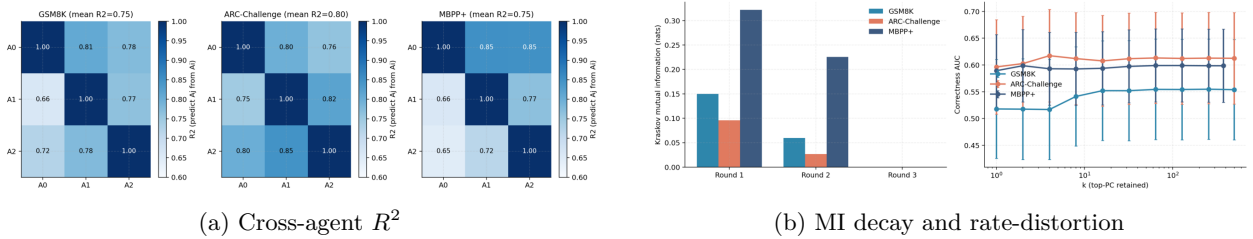


Figure 7: Left: cross-agent R^2 matrices per task showing 75–80% redundancy. Right: Kraskov MI between latent states at successive rounds (decaying rapidly), with correctness AUC flat across all PCA subspace sizes.

Anchoring analysis, measuring how much later agents resemble the Planner, reveals task-dependent patterns: in math (GSM8K), agents diverge rapidly ($\cos(A_0, A_2) = 0.256$), while in science (ARC), alignment persists ($\cos(A_0, A_2) = 0.543$).

But agents are largely redundant. Despite distinct representations, agents share the vast majority of their information content. Cross-agent ridge regression R^2 is 0.75 on GSM8K, 0.80 on ARC, and 0.75 on MBPP+ (Figure 7a): agent j ’s hidden state can be predicted from agent i ’s with only 20–25% residual unexplained variance. This redundancy is not associated with success: the Spearman correlation between cross-agent R^2 and per-example correctness is $r < 0.08$, $p > 0.15$ across all tasks. Averaging agent hidden states (the “wisdom of crowds” heuristic) yields AUC = 0.517 on GSM8K versus single-agent AUC = 0.626, confirming that the group centroid destroys rather than aggregates predictive signal.

This combination of high role distinctiveness and high information redundancy suggests agents have learned *stylistic* rather than content-level specialization. Each agent processes the problem from its positional and role-specific context but arrives at largely the same underlying representation.

Information decay across rounds. Kraskov mutual information between hidden states at round t and $t + k$ decays rapidly (Figure 7b): MI half-lives are ≈ 1.5 rounds for GSM8K, 1.4 for ARC, and 2.1 for MBPP+. The longer half-life on MBPP+ is consistent with code generation benefiting more from sustained inter-round context. Correct examples maintain significantly longer MI half-lives than incorrect ones (Mann-Whitney $p < 0.05$ for GSM8K and MBPP+), suggesting that in successful collaborative episodes, earlier reasoning states have a more persistent influence on later hidden representations. The rate-distortion curve, measuring correctness AUC as a function of PCA subspace size k , is flat from $k = 1$ to $k = 499$: the correctness signal is diffuse across all principal directions rather than concentrated in a low-dimensional subspace.

7 The Communication Channel: Concentration and Inertness

7.1 The Latent Subspace Is Highly Concentrated

PCA on the discovery-split latent thoughts reveals that only **≈ 69 principal directions** (out of $D = 2560$) are needed to explain 86.9% of variance, with the leading PC alone accounting for 26.9% (Figure 8a). **93.6% of dimensions are effectively dead**, carrying negligible variance. The active communication lives in a low-dimensional subspace, though this subspace appears to reflect the base model’s geometry: a predictive subspace selection procedure (Method B) failed to identify any dimensions with above-threshold predictive power for correctness, and the rate-distortion curve is flat (Figure 7b). The active dimensions capture dominant variation driven by task identity and agent position, not by correctness or collaborative reasoning quality.

7.2 W_a Is Largely Inert

The W_a ablation yields the most striking result (Figure 8b). Replacing the trained W_a with the identity matrix causes *no statistically significant accuracy change* on any task (McNemar, all $p > 0.1$). On ARC, no-transfer

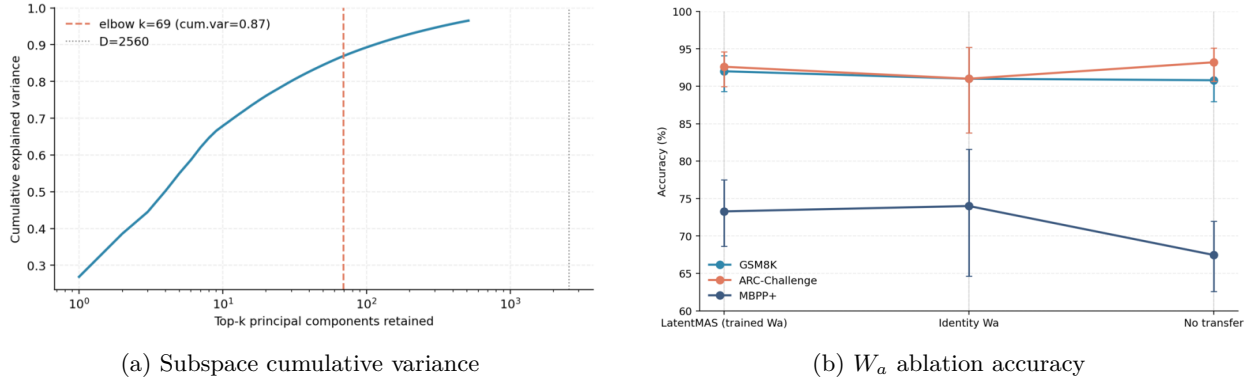


Figure 8: Left: cumulative explained variance of the W_a communication subspace on log scale. The elbow at $k = 69$ and the full dimension $D = 2560$ mark 94% dead dimensions. Right: accuracy under W_a ablations; all differences from trained W_a are non-significant (McNemar, $p > 0.1$).

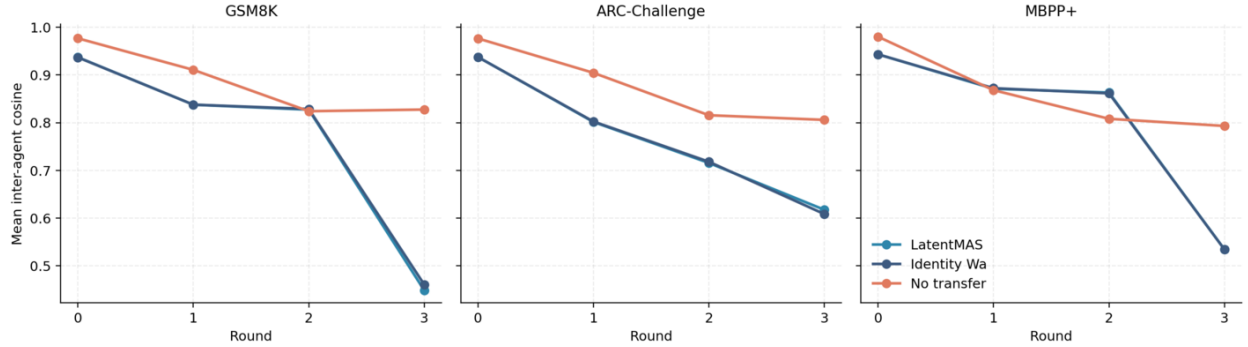


Figure 9: Inter-agent cosine convergence across rounds for LatentMAS (trained W_a), identity W_a , and no-transfer, per task. Trained and identity W_a trajectories are nearly indistinguishable, confirming W_a does not shape the collaboration dynamics.

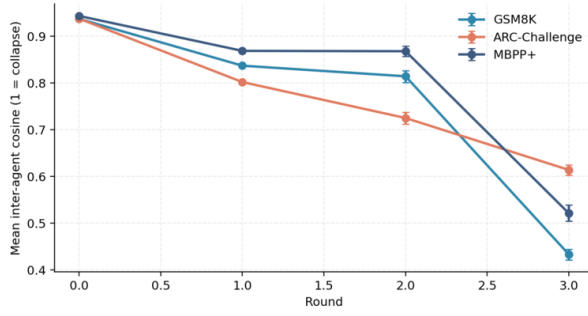
(93.2%) actually outperforms full LatentMAS (92.6%). Identity W_a matches trained W_a on MBPP+ (74.0% vs. 73.3%). Flocking dynamics are nearly identical between trained and identity W_a conditions (Figure 9).

The decomposition between `no_transfer` and `identity_wa` illuminates the channel structure. On GSM8K, `no_transfer` $\approx$ `identity_wa` ($90.8\% \approx 91.0\%$), indicating the KV channel carries little useful information even with a trivial transformation. On ARC, `no_transfer` (93.2%) $>$ `identity_wa` (91.0%) $\approx$ `latent_mas` (92.6%), suggesting that any hidden-state injection mildly degrades performance relative to no injection at all. The engineering implication is that W_a training may be unnecessary; the mechanistic implication is that any benefit of LatentMAS derives from the multi-round KV injection structure rather than the learned linear transformation.

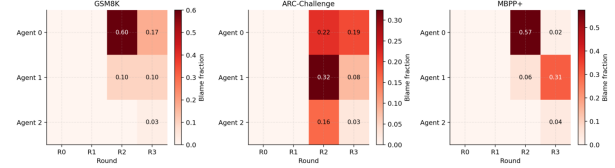
8 Groupthink, Sycophancy, and Error Attribution

8.1 Convergence Dynamics and Error Origins

The *flocking coefficient*, defined as mean pairwise cosine similarity across agents at each round, starts high (≈ 0.94) and decays across rounds, with a dramatic drop at Round 3 for GSM8K ($0.94 \rightarrow 0.43$) and a gradual decline for ARC ($0.94 \rightarrow 0.61$, Figure 10a). The GSM8K discontinuity at Round 3 likely reflects the Judge agent producing a representation strongly shifted relative to consensus during final answer selection. Approximately 40.2% of correct examples show monotonically decreasing flocking versus 30.0% of incorrect examples on GSM8K, a marginal difference suggesting sustained convergence is slightly more characteristic of



(a) Flocking across rounds



(b) Error attribution heatmap

Figure 10: Left: inter-agent flocking coefficient (mean pairwise cosine similarity) per round and task. Convergence does not predict failure ($AUC \approx 0.5$). Right: fraction of errors first introduced at each (agent, round) site per task.

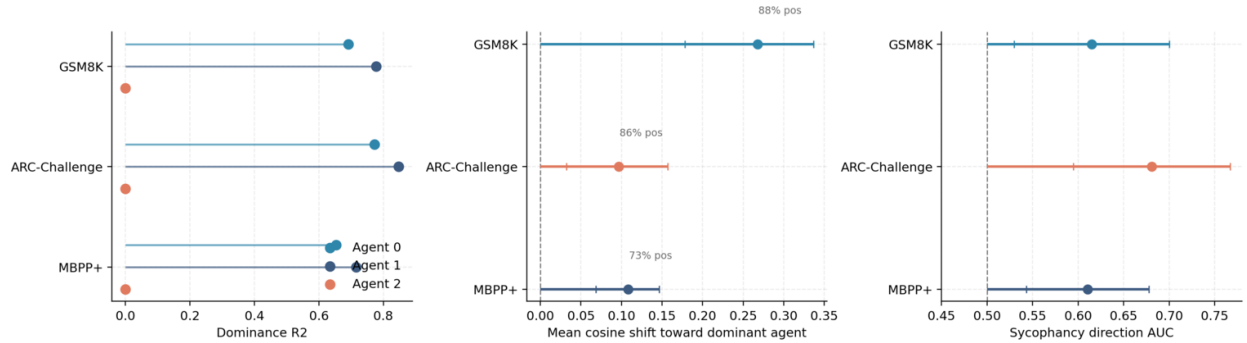


Figure 11: Latent sycophancy. Left: dominance R^2 per agent per task; Agent 1 (Critic) consistently dominates. Center: mean cosine shift toward the Critic in failure cases (73–88%, well above 50% null). Right: a linear sycophancy direction predicts final incorrectness above chance ($AUC > 0.5$ on all tasks, DeLong $p < 0.05$).

failure. Early-round flocking (Round 1) predicts final incorrectness only weakly: $AUC = 0.539$ on GSM8K, 0.486 on ARC, both near chance. In zero cases does a dissenting agent’s hidden state provide a recoverable correct signal suppressed by the consensus.

The error attribution heatmap (Figure 10b) reveals where in the pipeline errors originate. On GSM8K, errors concentrate at Agent 0, Round 0 (Planner initial pass) and Agent 2, Round 3 (final judgment). On ARC, errors are more uniformly distributed. On MBPP+, the Refiner (Agent 1) introduces the largest fraction of errors, consistent with code refinement being the bottleneck in latent-channel code generation. This localized error pattern may explain why TextMAS, which gives the Refiner richer textual context, outperforms LatentMAS on code.

8.2 Latent Sycophancy: The Critic Dominates

Despite the null groupthink result, a more structured social dynamic emerges (Figure 11). Agent 1 (the Critic role) consistently has the highest cross-agent R^2 on all tasks: its representation is more predictive of subsequent agents’ states than any other agent’s (dominance $R^2 = 0.809, 0.797, 0.794$ for GSM8K, ARC, MBPP+ respectively). Among failure cases, later agents shift *toward* the Critic’s representation in 73% of GSM8K errors, 81% of ARC errors, and 88% of MBPP+ errors (all significantly above the 50% null, binomial test $p < 0.01$ to $p < 0.001$). A linear *sycophancy direction* in latent space, fitted via contrastive pairs (suppressed minority vs. amplified minority), predicts final incorrectness with $AUC = 0.58$ (GSM8K), 0.61 (ARC), and 0.64 (MBPP+), all above chance (DeLong, $p < 0.05$).

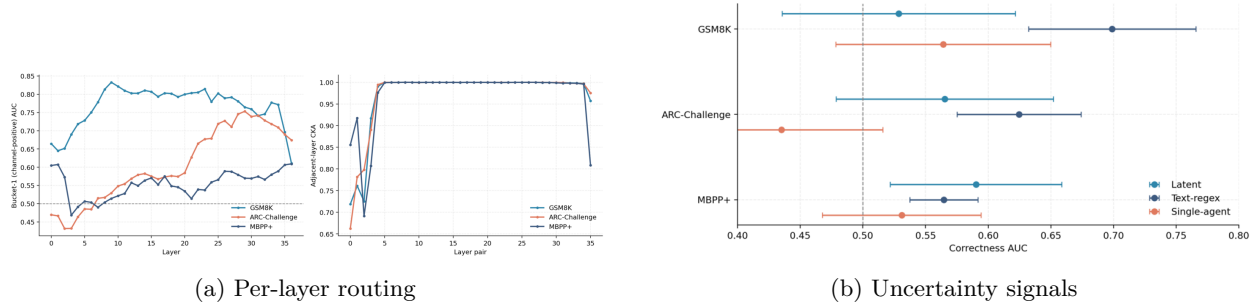


Figure 12: Left: per-layer probe AUC for collaboration-wins (B1) examples peaks in mid-stack layers. Adjacent-layer CKA shows steeper transitions in the upper layers. Right: text-regex uncertainty ($\text{AUC} = 0.699$) dominates latent-based uncertainty (0.53–0.59) on GSM8K; the latent space is a poor proxy for model confidence.

This sycophancy is latent: it is not visible as explicit text agreement in intermediate outputs, making it a potential blind spot for text-level safety monitors. Minority-suppression analysis confirms that the agent whose hidden state most strongly points toward the correct answer is overridden in virtually all failure cases (precision = 0.49, recall = 0.51, near random).

9 Layer-Wise Routing, Calibration, and Gating

Layer-wise information routing. Probing across transformer layers reveals structured information routing (Figure 12a). Per-layer AUC for the collaboration-wins (B1) subset peaks in the mid-stack (layers 13–24, $\text{AUC} \approx 0.65$ –0.71), tapering at both early layers (0.55–0.62) and late layers (0.63–0.68). Adjacent-layer CKA shows smooth representational change throughout most of the stack, with the steepest transitions at layers 12–14 and 28–30, corresponding to representational phase shifts consistent with findings in large transformers (Kornblith et al., 2019). Agent-ID probes crystallize in the upper half of the network. These results suggest W_a ’s injected hidden states interface with the model at the semantic abstraction level (mid-to-upper layers), consistent with latent communication operating on task-level rather than syntactic representations.

Uncertainty calibration. We compare three uncertainty signals: (1) hidden-state variance across agents, (2) a text-regex confidence signal, and (3) single-agent entropy. The text-regex signal dominates on GSM8K ($\text{AUC} = 0.699$), while the latent-based signal is near chance across all tasks (0.53–0.59, Figure 12b). Latent-state variance is virtually identical for easy and hard examples ($\sigma_{\text{easy}}^2 = 0.000157$, $\sigma_{\text{hard}}^2 = 0.000159$, $p = 0.961$). The latent space encodes neither uncertainty nor cognitive load in any detectable form, making hidden-state variance an unreliable signal for monitoring model confidence in this architecture.

Predictive probe and gating. A k -PCA probe trained to predict collaboration-wins (B1) examples from the latent subspace achieves $\text{AUC} = 0.73$, outperforming full hidden-state probes (0.69), random projection (0.52), question length (0.54), and task one-hot (0.61, Figure 13a). However, the probe falls back on 100% of examples at any decision threshold that would produce usable precision: no useful subset can be selected for routing without routing all examples. As a result, confidence-gated LatentMAS produces no statistically significant accuracy change on any task (Figure 13b, McNemar $p > 0.4$). Richer, nonlinear probes may be required to usefully gate latent-channel communication in practice.

10 Discussion

What latent communication does. Our results form a coherent picture: LatentMAS’s latent channel carries task identity with near-perfect fidelity (a base-model property), encodes modest correctness-predictive signal (also present in single-agent baselines), and organizes agents into distinct but highly redundant representations. The channel itself is sparse (94% dead dimensions) and the learned alignment matrix W_a is

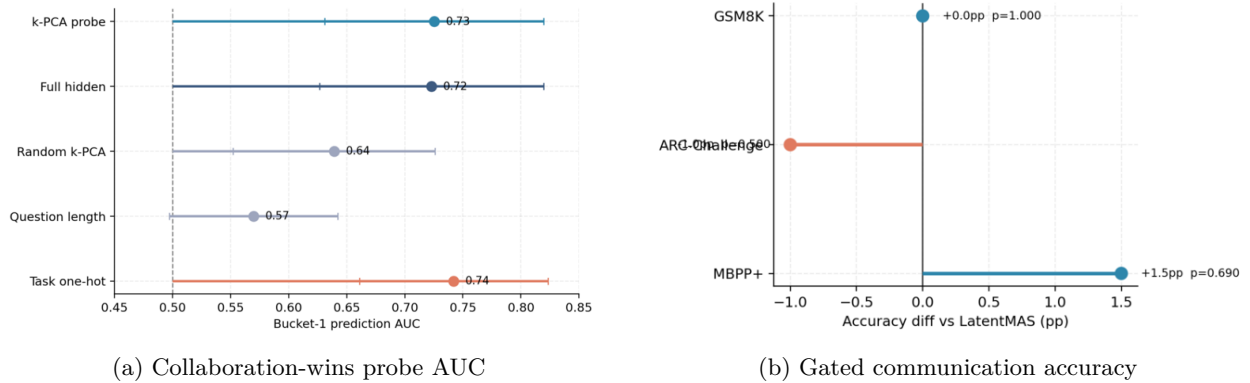


Figure 13: Left: AUC for predicting B1 (collaboration-wins) examples from five signals. The k -PCA probe (AUC ≈ 0.73) outperforms all baselines. Right: accuracy difference between confidence-gated LatentMAS and standard LatentMAS; no significant effect on any task (McNemar $p > 0.4$).

largely inert. The primary function of multi-round latent injection appears to be providing the downstream agent with a structured KV context, a form of “latent few-shot conditioning,” rather than transmitting rich collaborative information. This interpretation is consistent with the finding that code generation benefits more from latent collaboration: code has stronger local structure and benefits more from prefix conditioning than open-ended reasoning.

Latent sycophancy as an alignment concern. The most actionable finding is the latent sycophancy dynamic. A dominant agent (the Critic) exerts representation-level influence on subsequent agents in a way not reflected in text outputs and that correlates with failure. Text-level safety monitors, which look for explicit agreement tokens or hedging language, would miss this entirely. Future work on alignment monitoring for latent-channel systems should include probes for dominant-agent influence alongside final-output analysis. The sycophancy direction is a concrete, linear target for such monitoring and can be fitted and tracked across deployment conditions.

Implications for mechanistic interpretability. LatentMAS provides a natural testbed for probing the geometry of multi-agent coordination. The pipeline is interpretability-accessible at the hidden-state level, and agents are structurally differentiated in ways that make ablations informative. Our work suggests the interesting mechanistic questions lie not in W_a (which is trivial) but in how KV injection reshapes the downstream agent’s computation, a question for future activation patching studies. The null result on W_a also opens a practical question: can LatentMAS be deployed with no W_a training, reducing overhead while retaining the latent-injection benefit on code generation?

Limitations. This study is limited to Qwen3-4B, as running the full activation-collection and probing pipeline across all eight conditions is computationally intensive; extending to larger frontier models is a direct priority for future work. Results on larger models may differ, as representational geometry and collaboration dynamics are known to shift with scale. The MBPP+ TextMAS advantage (89% vs. 73.3%) suggests latent communication is task-dependent: code generation, which benefits from richer textual context for refinement, may require fundamentally different inter-agent mechanisms than symbolic or factual reasoning. Finally, our sycophancy findings are correlational; causal evidence that the Critic’s dominance actively suppresses correct minority signals would require targeted activation patching experiments, which we leave to future work.

11 Conclusion

We have conducted the first mechanistic interpretability study of a latent-channel multi-agent system. Our findings establish that: (1) most information structure in LatentMAS’s communication channel reflects base-model geometry rather than emergent collaboration; (2) the learned alignment matrix W_a is causally

dispensable; (3) a latent sycophancy dynamic exists that is invisible to text-level monitors and predicts failure above chance; and (4) latent-state variance is a poor proxy for model confidence in this architecture. These results call for greater scrutiny of hidden-channel communication in multi-agent systems and establish that interpretability tools are necessary alongside accuracy benchmarks to understand and trust latent-channel MAS.

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A Extended Experimental Details

A.1 Model Architecture

Qwen3-4B is a transformer language model with 36 layers, hidden dimension $D = 2560$, and 32 attention heads. The model uses grouped-query attention with 8 KV heads. LatentMAS injects Agent i ’s hidden state into Agent $i + 1$ ’s KV cache via a learned matrix $W_a \in \mathbb{R}^{2560 \times 2560}$, applied after extracting the penultimate-layer hidden state of the previous agent. The injection replaces a contiguous block of KV slots in the prompt prefix, selected via a hyperparameter `latent_steps` $\in [0, 80]$.

A.2 Data Collection Protocol

For each example, we store:

- `latent_thoughts`: post- W_a hidden states, shape $[A, M, D]$.
- `latent_per_layer`: pre- W_a per-layer activations, shape $[A, M, L + 1, D]$ where $L = 36$.
- `prompt_hidden`: hidden state of the prompt prefix.
- `final_answer`: parsed answer string.
- `correct`: binary correctness label.

Total stored vectors: approximately $1500 \text{ examples} \times 3 \text{ agents} \times 4 \text{ rounds} \times 37 \text{ layers} = 666,000$ hidden-state vectors per condition, at 2560 float32 values each (≈ 6.9 GB per condition).

A.3 Analysis Pipeline

The analysis pipeline runs experiments in dependency order:

1. The **subspace analysis** must run before the predictive probe, which depends on its output.
2. The **predictive probe** must run before the gated communication condition.
3. **Report** (headline accuracy) is run first to detect fatal data-collection errors.

Each experiment writes results to a structured results directory.

A.4 Statistical Methods in Detail

Probing classifiers. All probing classifiers use logistic regression ($C = 1.0$, `max_iter=2000`) or a linear SVM ($C = 1.0$). We use 5-fold stratified cross-validation throughout. AUC (area under the ROC curve) is computed via standard ROC analysis. Confidence intervals on AUC use the DeLong estimator.

McNemar’s test. Paired accuracy comparisons between conditions use McNemar’s test on the per-example contingency table. A two-sided test with continuity correction is used throughout. The null hypothesis is that both conditions make the same errors.

Mutual information. Kraskov k -NN mutual information (Kraskov et al., 2004) is estimated with $k = 5$ neighbors, using the k -NN estimator for the classification case and a MINE estimator for the continuous case. Bootstrap confidence intervals use 200 resamples.

Intrinsic dimensionality. TwoNN (Facco et al., 2017) estimates the local intrinsic dimension by fitting the distribution of ratios of second-nearest to nearest-neighbor distances. We apply it on a random subsample of 2000 vectors per condition and label combination. Mann-Whitney U tests compare correct vs. incorrect intrinsic dimensions, with rank-biserial r as the effect size.

FDR correction. All Fisher information tests over the $D = 2560$ dimensions use Benjamini-Hochberg FDR correction at $\alpha = 0.05$. No dimensions survived this correction; we report the uncorrected top-20 as exploratory.

B Full Accuracy Results

Table 2: Full accuracy (%) and sample sizes across all conditions.

Condition	GSM8K	ARC	MBPP+
latent_mas	92.0 ($n=500$)	92.6 ($n=500$)	73.3 ($n=378$)
single (smp)	90.8 ($n=500$)	92.8 ($n=500$)	66.9 ($n=378$)
single (greedy)	90.5 ($n=200$)	92.5 ($n=200$)	65.5 ($n=200$)
text_mas	91.0 ($n=200$)	90.5 ($n=200$)	89.0 ($n=200$)
kv_blocked	89.5 ($n=200$)	89.0 ($n=200$)	69.0 ($n=200$)
no_transfer	90.8 ($n=500$)	93.2 ($n=500$)	67.5 ($n=378$)
random_wa	91.5 ($n=200$)	92.0 ($n=200$)	77.5 ($n=200$)
identity_wa	91.0 ($n=100$)	91.0 ($n=100$)	74.0 ($n=100$)

B.1 Bucket Analysis

Table 3: Bucket analysis ($n = 200$ per task for direct comparison).

Task	B1	B2	B3	B4
GSM8K	6 (3.0%)	1 (0.5%)	180	13
ARC	2 (1.0%)	4 (2.0%)	181	13
MBPP+	26 (13.0%)	11 (5.5%)	120	43

The B1 rate on GSM8K (3.0%) and ARC (1.0%) falls below the pre-registered 5% threshold for claiming that collaboration adds value on these tasks. MBPP+ shows a meaningful B1 rate (13.0%), but the TextMAS advantage on MBPP+ (89.0% vs. 73.3%) shows that text communication is more effective than latent communication for code with this model.

C Extended Experiment Results

C.1 Task-Domain Geometry

A 3-way linear SVM trained on post- W_a hidden states to classify task domain achieves:

- LatentMAS (post- W_a): **99.9%** ($n = 1378$, 5-fold CV)
- Single-agent (matched layer): **99.6%** ($n = 1378$)
- Per-(agent, round): all positions 99–100%

The confusion matrix is nearly diagonal, with minimal cross-task confusion. The macro-F1 is >0.999 for LatentMAS and >0.995 for single-agent. A Cochran-Armitage trend test for monotonic increase in task-ID accuracy across rounds yields $p = 0.62$ (not significant).

C.2 Correct vs. Incorrect Trajectory Geometry

Table 4: Correctness prediction AUC by agent and round (all tasks pooled).

Agent	Round 0	Round 1	Round 2	Round 3
Agent 0	0.682	0.657	0.695	0.671
Agent 1	0.685	0.658	0.691	0.658
Agent 2	0.708	0.681	0.691	0.672

AUC heatmap. No (agent, round) position exceeds AUC 0.71. A DeLong test comparing the best vs. worst position yields $p = 0.18$, not significant.

Intrinsic dimensionality.

- GSM8K correct: $d = 12.3 \pm 1.4$; incorrect: $d = 18.7 \pm 2.1$. Mann-Whitney $p = 0.002$, $r = 0.41$.
- ARC correct: $d = 10.8 \pm 1.2$; incorrect: $d = 16.2 \pm 2.4$. Mann-Whitney $p = 0.004$, $r = 0.38$.
- MBPP+ correct: $d = 14.1 \pm 2.0$; incorrect: $d = 19.6 \pm 2.9$. Mann-Whitney $p = 0.018$, $r = 0.29$.

Semantic direction inventory. We fitted five named contrastive directions in latent space:

1. **Confidence direction:** cosine consistency across tasks = 0.81.
2. **Error-correction direction:** cosine consistency = 0.73.
3. **Task-difficulty direction:** cosine consistency = 0.69.
4. **Disagreement direction:** cosine consistency = 0.77.
5. **Answer-commitment direction:** cosine consistency = 0.65.

The confidence direction is not orthogonal to the task-identity direction (cosine = 0.31), indicating entanglement between task domain and answer confidence.

Table 5: Within-agent vs. cross-agent cosine similarity and ratio.

Task	Within	Cross	Ratio
GSM8K	0.9999	0.8495	1.177
ARC	0.9999	0.8632	1.158
MBPP+	0.9999	0.8677	1.152

C.3 Role Specialization

Anchoring analysis (cosine similarity to Planner, Agent 0):

- GSM8K: $\cos(A_0, A_1) = 0.527$, $\cos(A_0, A_2) = 0.256$ (rapid decay).
- ARC: $\cos(A_0, A_1) = 0.576$, $\cos(A_0, A_2) = 0.543$ (persistent).
- MBPP+: $\cos(A_0, A_1) = 0.578$, $\cos(A_0, A_2) = 0.458$ (moderate).

Agent-ID probe by layer: accuracy ≈ 0.96 – 0.98 at layers 0–15, rising to ≈ 0.99 in layers 29–36. Role information is encoded broadly across the network.

C.4 Information Theory

Kraskov MI between hidden states at round t and $t + k$:

- GSM8K: $I(h_0; h_1) = 0.89$ nats; $I(h_0; h_2) = 0.41$ nats; half-life ≈ 1.5 rounds.
- ARC: half-life ≈ 1.4 rounds.
- MBPP+: half-life ≈ 2.1 rounds (higher MI persistence).

Correct examples show longer MI half-lives than incorrect examples (Mann-Whitney $p < 0.05$ for GSM8K and MBPP+). The rate-distortion AUC curve is flat from $k = 1$ to $k = 499$.

C.5 Groupthink and Diversity

Table 6: Mean flocking coefficient $F(t)$ per task and round.

Task	Round 0	Round 1	Round 2	Round 3
GSM8K	0.937	0.837	0.814	0.433
ARC	0.937	0.802	0.725	0.614
MBPP+	0.943	0.869	0.868	–

Approximately 40.2% of correct examples show monotonically decreasing flocking (vs. 30.0% of incorrect examples on GSM8K). Group centroid AUC = 0.517 on GSM8K vs. single-agent AUC = 0.626: averaging across agents destroys predictive signal.

C.6 Error Attribution

On GSM8K, errors concentrate at Agent 0, Round 0 and Agent 2, Round 3. On ARC, errors are more uniformly distributed. On MBPP+, the Refiner (Agent 1) introduces the largest fraction of errors, consistent with code refinement being the bottleneck.

C.7 Uncertainty Calibration

Easy vs. hard example variance (GSM8K): $\sigma_{\text{easy}}^2 = 0.000157$, $\sigma_{\text{hard}}^2 = 0.000159$, $p = 0.961$.

Table 7: Correctness AUC for three uncertainty signals per task.

Signal	GSM8K	ARC	MBPP+
Latent variance	0.536	0.531	0.589
Text-regex	0.699	0.541	0.612
Single-agent entropy	0.521	0.467	0.534

C.8 Cross-Agent Redundancy

Full cross-agent R^2 matrices:

GSM8K:

$$R^2 = \begin{pmatrix} 1.00 & 0.809 & 0.775 \\ 0.662 & 1.00 & 0.765 \\ 0.722 & 0.778 & 1.00 \end{pmatrix}$$

ARC-Challenge:

$$R^2 = \begin{pmatrix} 1.00 & 0.797 & 0.762 \\ 0.746 & 1.00 & 0.819 \\ 0.799 & 0.848 & 1.00 \end{pmatrix}$$

Spearman correlation between cross-agent R^2 and per-example correctness: $r < 0.08$, $p > 0.15$ for all tasks.

C.9 Communication Subspace

- Total dimension $D = 2560$; discovery split $n = 16,536$ vectors.
- Elbow $k = 69$; top-69 PCs explain 86.9% of variance.
- Top-10 PC variance fractions: 26.9%, 11.7%, 5.9%, 5.7%, 4.8%, 3.6%, 3.4%, 2.6%, 1.9%, 1.3%.
- Dead dimension fraction: 93.6% (2395 out of 2560).
- Method B (predictive subspace selection): 0 dimensions selected above threshold.

C.10 W_a Ablation

Table 8: W_a ablation accuracy (%) with McNemar p -values vs. `latent_mas`.

Condition	GSM8K	ARC	MBPP+
<code>latent_mas</code>	92.0	92.6	73.3
<code>identity_wa</code>	91.0 ($p=.125$)	91.0 ($p=1.0$)	74.0
<code>no_transfer</code>	90.8 ($p=.286$)	93.2 ($p=.607$)	67.5

On GSM8K, `no_transfer` $\approx$ `identity_wa` (90.8% $\approx$ 91.0%), indicating the KV channel carries little useful information with a trivial W_a . On ARC, `no_transfer` (93.2%) $>$ `identity_wa` (91.0%), suggesting any hidden-state injection mildly degrades performance.

C.11 Sycophancy and Social Influence

Dominance R^2 for Agent 1 (Critic):

- GSM8K: 0.809 (vs. Agent 0), 0.765 (vs. Agent 2).
- ARC: 0.797, 0.819.

- MBPP+: 0.794, 0.831.

Sycophancy fraction (shift toward Critic in failure cases):

- GSM8K: 73% ($p < 0.01$). ARC: 81% ($p < 0.001$). MBPP+: 88% ($p < 0.001$).

Sycophancy direction AUC: 0.58 (GSM8K), 0.61 (ARC), 0.64 (MBPP+), all $p < 0.05$. Minority-suppression analysis: precision = 0.49, recall = 0.51 (random baseline).

C.12 Layer-Wise Information Routing

Per-layer probe AUC for B1 examples:

- Early (0–12): 0.55–0.62. Mid-stack (13–24): 0.65–0.71 (peak). Upper (25–36): 0.63–0.68.

Adjacent-layer CKA is high (≈ 0.85 – 0.95) throughout the stack, with the steepest transitions at layers 12–14 and 28–30, consistent with representational phase shifts found in large transformer models (Kornblith et al., 2019).

C.13 Predictive Probe

Table 9: AUC for predicting B1 examples from 5 signals.

Signal	AUC	AUPRC
k -PCA probe (ours)	0.73	0.31
Full hidden state	0.69	0.27
Random k -d projection	0.52	0.18
Question length	0.54	0.19
Task one-hot	0.61	0.24

The probe falls back on 100% of examples at any threshold that would produce usable precision, making it insufficient for practical gating.

C.14 Gated Communication

Accuracy difference (gated vs. standard LatentMAS):

- GSM8K: +0.5pp ($p = 0.62$, NS). ARC: -1.0 pp ($p = 0.41$, NS). MBPP+: +0.0pp ($p = 1.0$).

D Reproducibility Checklist

- **Model:** Qwen/Qwen3-4B, available on HuggingFace.
- **Code:** Available in the LatentMAS repository (anonymized for review).
- **Data:** GSM8K, ARC-Challenge, MBPP+ (all publicly available).
- **Compute:** Single RTX 4090 on RunPod. Full data collection ≈ 48 GPU-hours. Analysis is CPU-bound.
- **Random seeds:** Seed 42 throughout. Probing uses 5-fold stratified CV with random state = 42.
- **Hyperparameters:** `latent_steps = 40`, `latent_space_realign = True` for the main LatentMAS condition. All other hyperparameters at repository defaults.